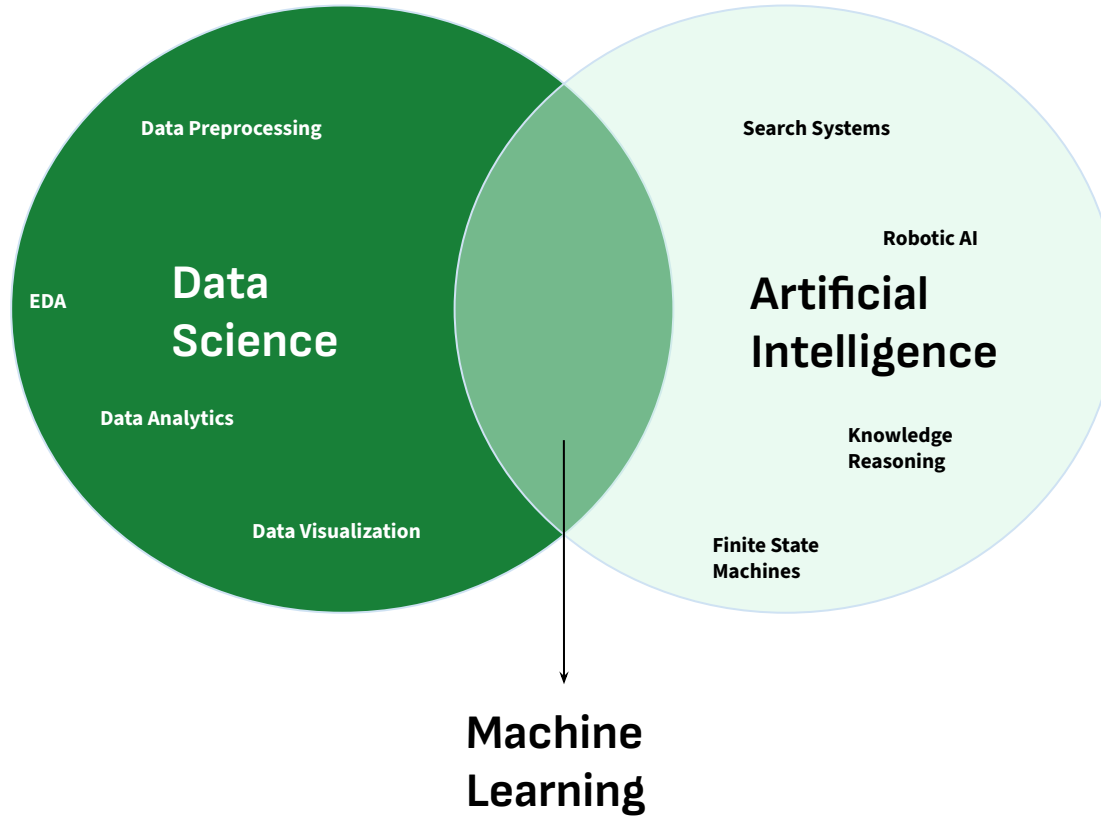


# AI vs DS vs ML vs DL vs Gen AI vs Agentic AI

Gen AI Glossary for Beginners

## Data Science vs Artificial Intelligence



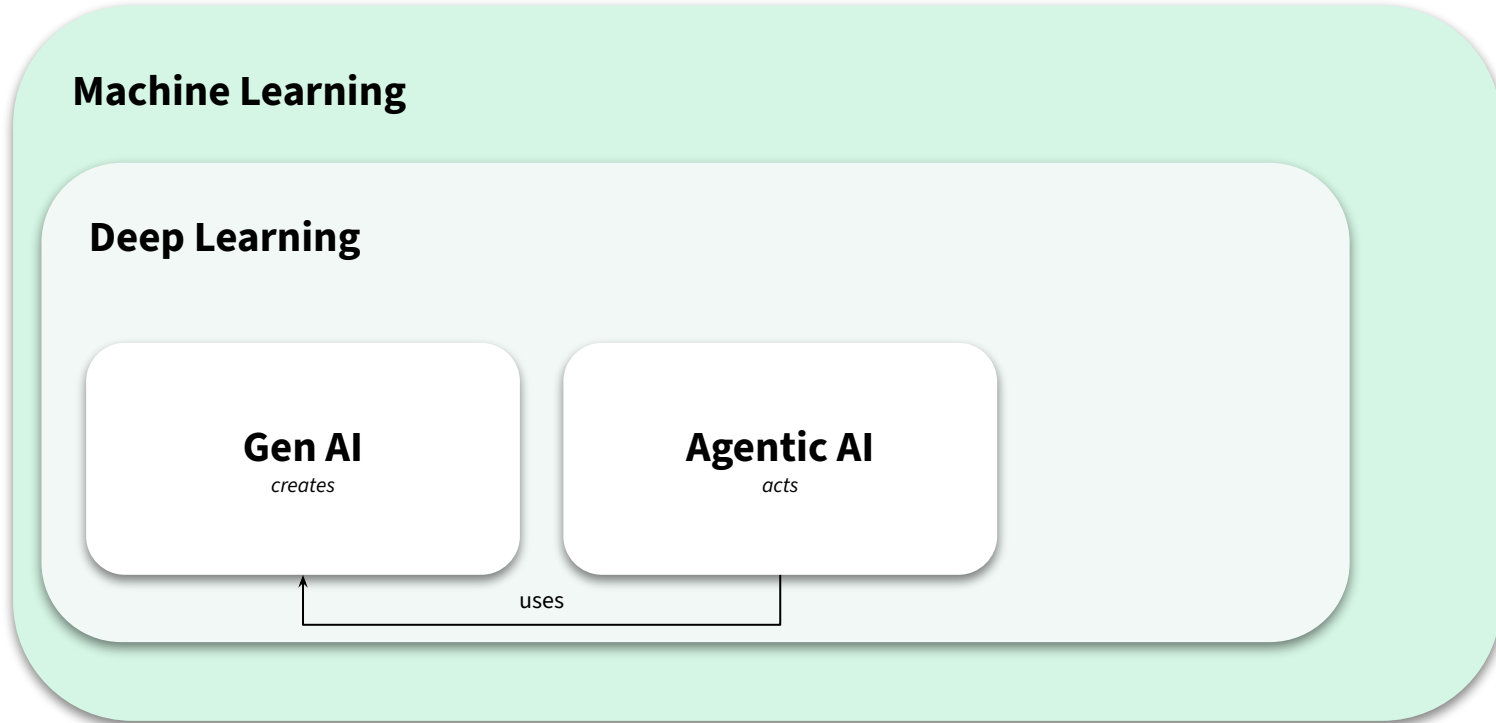


## What is Machine Learning?

Machine Learning is the science of converting human intuition into algorithms and code.



## ML vs DL vs Gen AI vs Agentic AI.



# What are Foundation Models?

Gen AI Glossary for Beginners



## Multiple Domains

Audio

Image & Video

Text

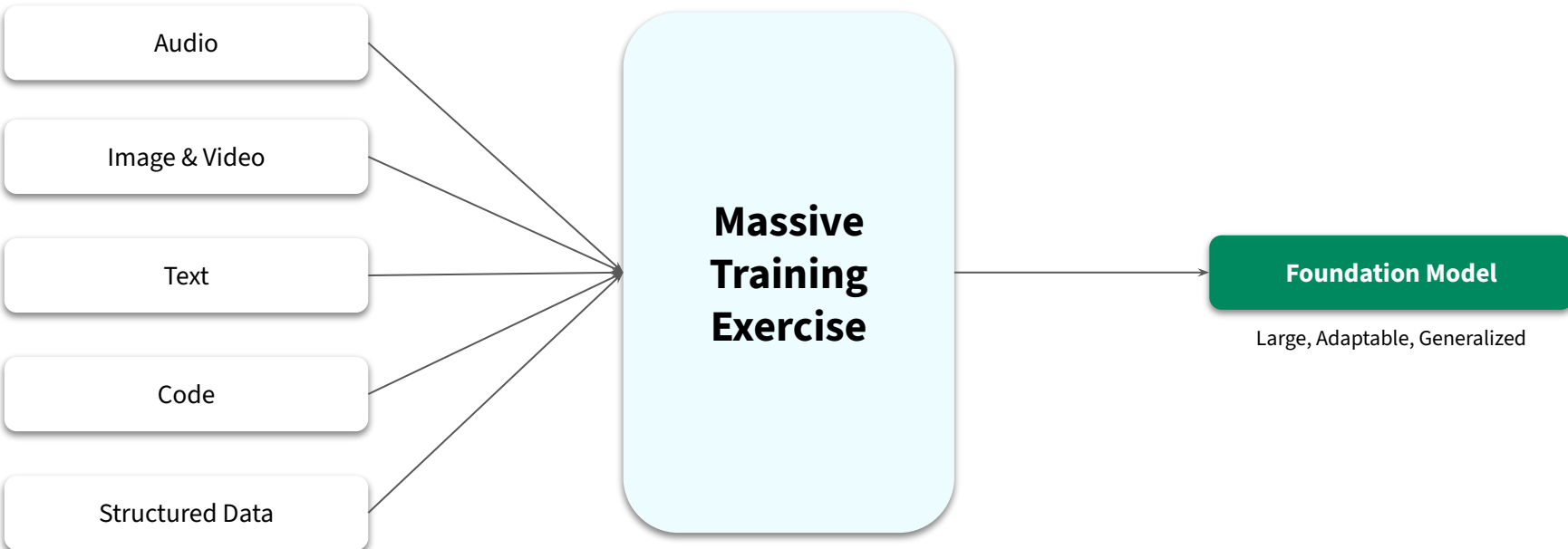
Code

Structured Data

**Massive  
Training  
Exercise**

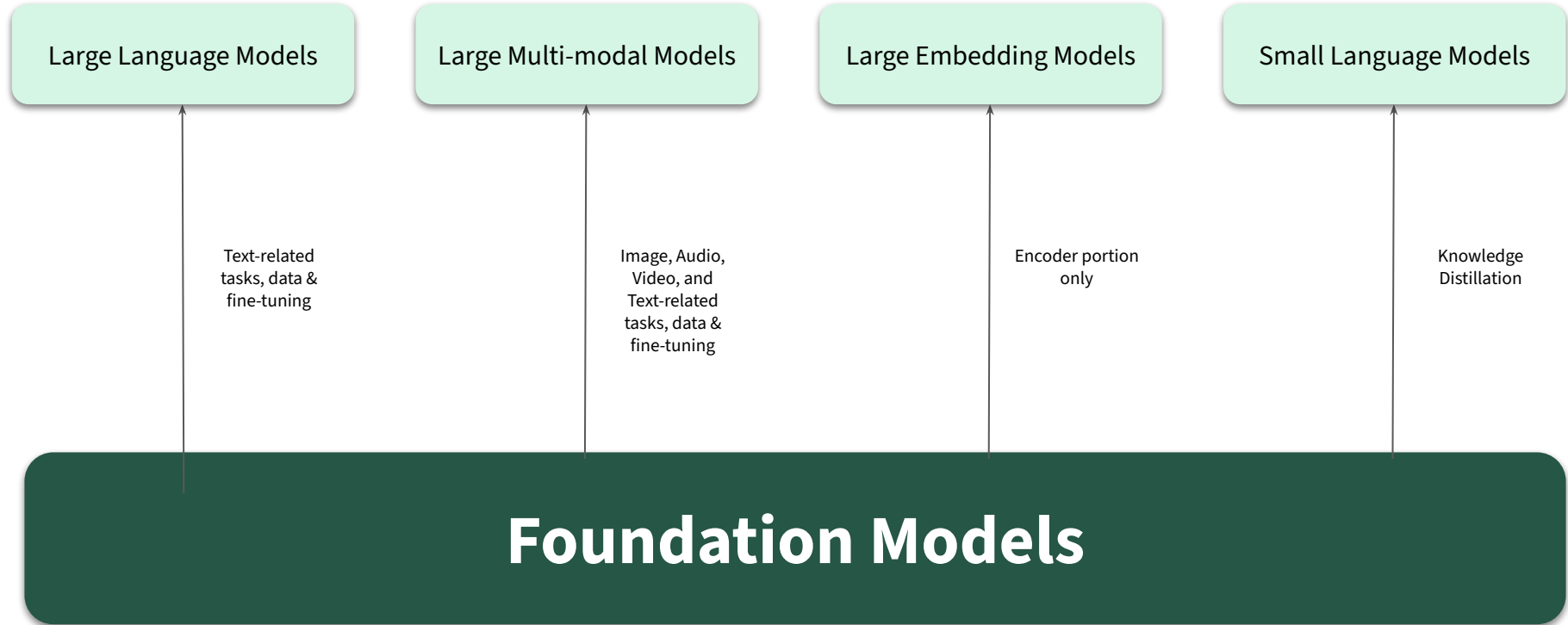
**Foundation Model**

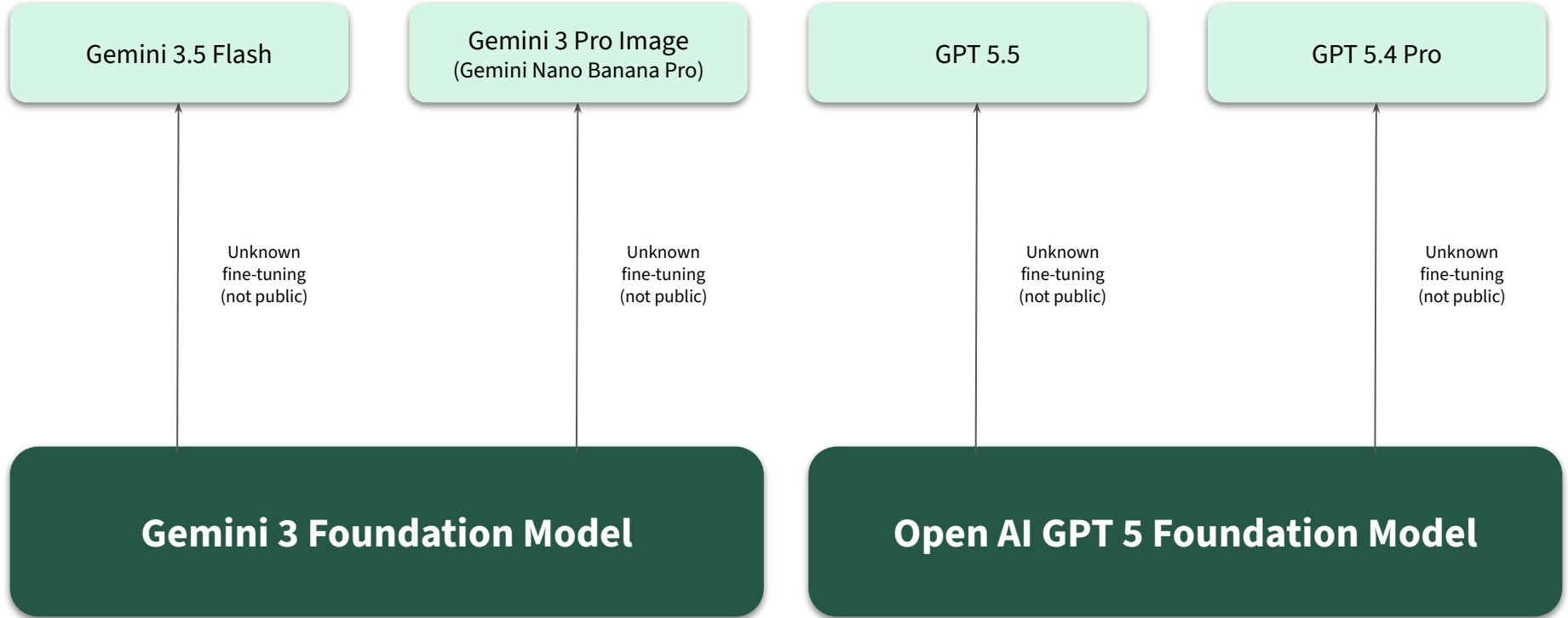
Large, Adaptable, Generalized





## LLMs: A Powerful Adaptation of Foundation Models







# What are SLMs?

Gen AI Glossary for Beginners



# Small Language Models: Are they the future of Gen AI in Enterprise?

SLMs

**Large Language Models**

# Anatomy of an Interaction with an LLM

Gen AI Glossary for Beginners

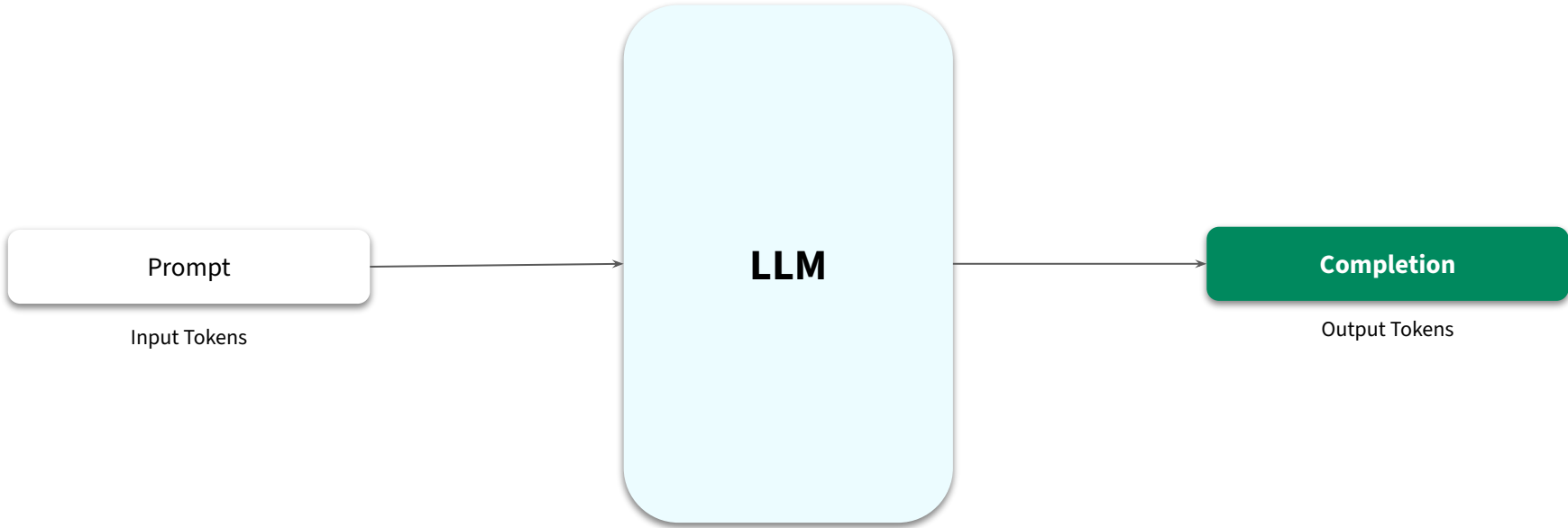


Prompt

**LLM**

**Completion**







## Context Window: The Model's Memory

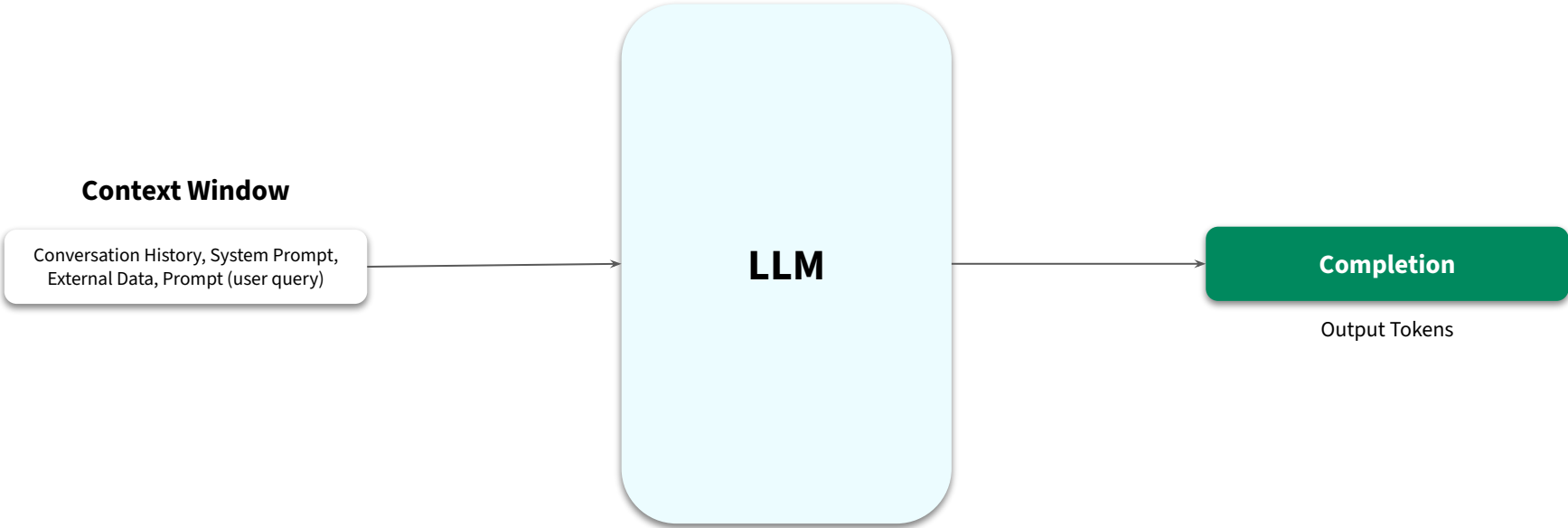
### Context Window

Conversation History, System Prompt,  
External Data, Prompt (user query)

**LLM**

**Completion**

Output Tokens



# Steering your LLM with settings

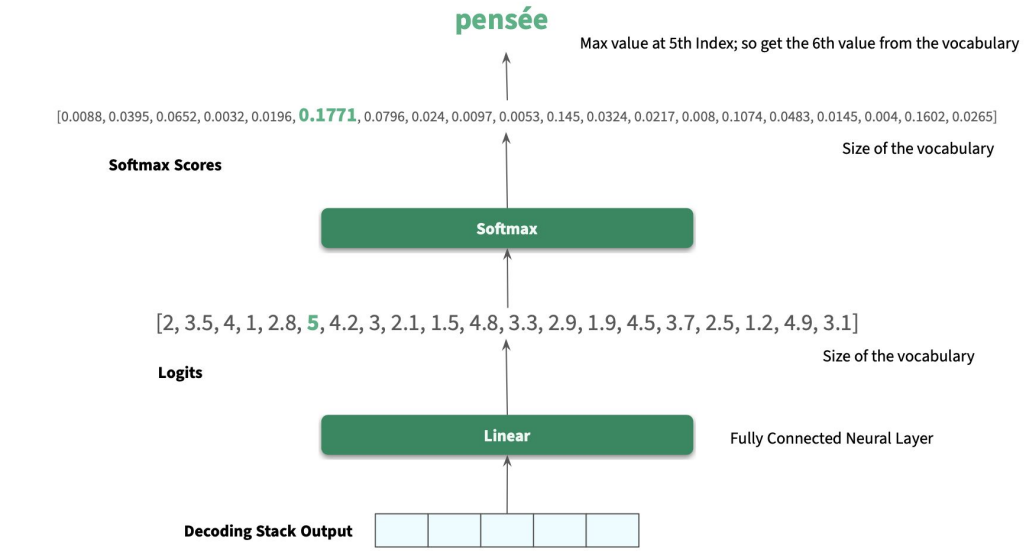
Gen AI Glossary for Beginners

What do settings control?

# Probability Distribution

the softmax layer outputs

*Or sometimes, directly, the logits*







## The best way to start the day is with a hot cup of...

coffee: 65%

tea: 25%

water: 5%

tears: 4%

cup: 0.2%

and so on.. for each token in the vocabulary.



## The best way to start the day is with a hot cup of...

coffee: 65%

tea: 25%

water: 5%

tears: 4%

cup: 0.2%

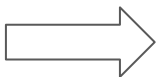
and so on.. for each token in the vocabulary.

**top\_k = 3**

coffee: 65%

tea: 25%

water: 5%



*renormalization*

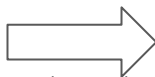
**top\_k = 3**

coffee: 68.4%

tea: 26.3%

water: 5.2%

*sums to one*



*random number  
between 0 to 1.0 is  
generated*

**top\_k = 3**

coffee will be selected if random value is between 0.00 to 0.684

tea will be selected if random value is between 0.685 to 0.947

water will be selected if random value is between 0.948 to 1.0



## The best way to start the day is with a hot cup of...

coffee: 65%

tea: 25%

water: 5%

tears: 4%

cup: 0.2%

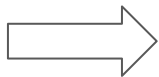
and so on.. for each token in the vocabulary.

---

**top\_p = 0.8**

coffee: 65%

tea: 25%



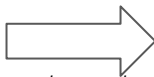
*renormalization*

**top\_p = 0.8**

coffee: 72.2%

tea: 27.8%

*sums to one*



*random number  
between 0 to 1.0 is  
generated*

**top\_p = 0.8**

coffee will be selected if random value is between 0.00 to 0.722

tea will be selected if random value is between 0.723 to 1.0



# The best way to start the day is with a hot cup of...

coffee: 65%

tea: 25%

water: 5%

tears: 4%

cup: 0.2%

and so on.. for each token in the vocabulary.

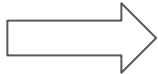
---

## original

coffee: 65%

tea: 25%

water: 5%



*Logits are divided  
by 0.2 before being  
sent to the softmax  
function*

## temperature = 0.2

coffee: 95%

tea: 5%

water: 0.1%

# The best way to start the day is with a hot cup of...

coffee: 65%

tea: 25%

water: 5%

tears: 4%

cup: 0.2%

and so on.. for each token in the vocabulary.

## original

coffee: 65%

tea: 25%

water: 5%



*Logits are divided  
by 1.8 before being  
sent to the softmax  
function*

## temperature = 1.8

coffee: 42%

tea: 40%

water: 14%



# The best way to start the day is with a hot cup of...

coffee: 65%

tea: 25%

water: 5%

tears: 4%

cup: 0.2%

and so on.. for each token in the vocabulary.

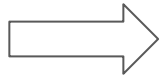
---

## original

coffee: 65%

tea: 25%

water: 5%



*Logits are divided  
by 1.0 before being  
sent to the softmax  
function; so no  
change*

## temperature = 1.0

coffee: 65%

tea: 25%

water: 5%



# The best way to start the day is with a hot cup of...

coffee: 65%

tea: 25%

water: 5%

tears: 4%

cup: 0.2%

and so on.. for each token in the vocabulary.

---

## original

coffee: 65%

tea: 25%

water: 5%

tears: 4%

cup: 0.2%



*Tokens already  
generated or in the  
prompt are heavily  
penalized*

## frequency\_penalty = 1.0

coffee: 65%

tea: 25%

water: 5%

tears: 4%

**cup: 0.00001%**



## A simple settings guide

| Use Case                         | Recommended Temperature | Recommended Top-P | Recommended Frequency Penalty | Rationale                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|-------------------------|-------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Factual Q&A / Summarization      | Low (0.2 - 0.5)         | High Top-P (0.9)  | 0.7 - 1.2                     | Prioritizes accuracy and determinism. Low temperature ensures the model sticks to the most likely, factual tokens. Penalties should be high to prevent repeating facts and to encourage the model to cover different points concisely.                                                                                          |
| Code Generation                  | Low (0.2 - 0.4)         | High Top-P (0.9)  | 0.1 - 0.4                     | Code requires precision and adherence to syntax. Low temperature prevents creative but non-functional code. Penalties must be low as code requires repeating variables and keywords. A high penalty could punish necessary syntax.                                                                                              |
| General Balanced Chat            | Medium (0.7 - 0.8)      | High Top-P (0.9)  | 0.3 - 0.6                     | A balance between being predictable and being engagingly creative, suitable for conversational agents. Penalties are moderate. Frequency penalty stops robotic phrases, while presence penalty encourages the conversation to move to new topics.                                                                               |
| Creative Writing / Brainstorming | High (0.9 - 1.2)        | Lower Top-P (0.8) | 0.0 - 0.4                     | Maximizes creativity and diversity of output. High temperature encourages novel ideas, while a slightly lower Top-P can prevent it from becoming completely incoherent. Penalties should be low. This allows for stylistic repetition (like a chorus), while the presence penalty gently nudges brainstorming toward new ideas. |